

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

Instructor: Ijaz Ul Haq, Ph.D.

ijazhaq@umich.edu

Location: MSB 104

Previous class:

- ✓ *What is data mining?*
- ✓ *Data mining vs Statistics vs Machine Learning*
- ✓ *The four big tasks*
- ✓ *CRISP-DM: the process we follow all semester*

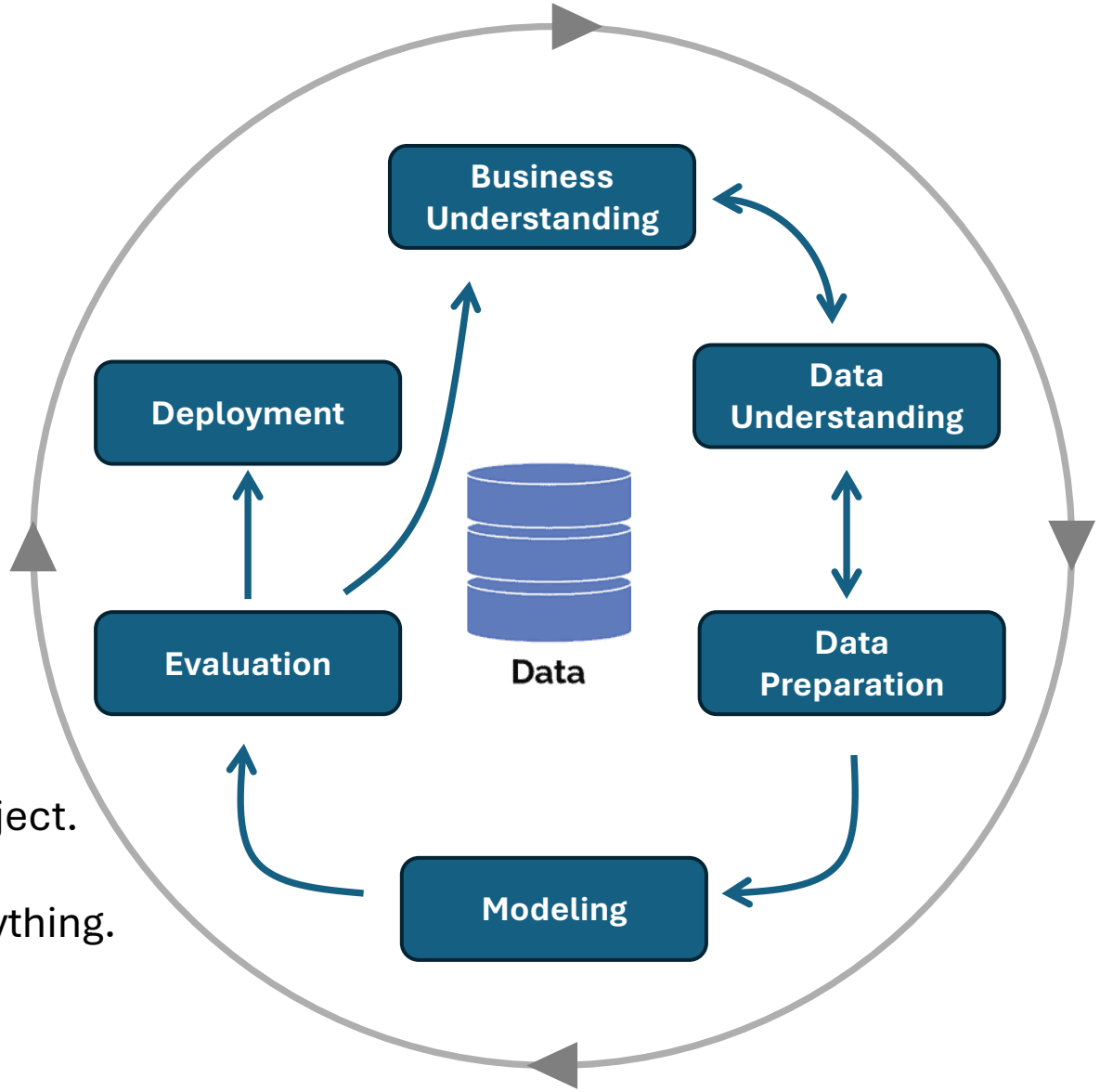
Today's class:

- *Alternatives to SCRIPS-DM Process Model (KDD and SEMMA)*
- *Our Working Environment*
- *Understanding Data (Data Attribution)*
- *EDA: finding problems, not making art*
- *Hands-on blocks in Colab*

CRISP-DM: How We Will Work

CRoss
Industry
Standard
Process for
Data Mining

- CRISP-DM is a process model for a data mining project.
- The algorithms live in step 4.
- The other five steps are what make step 4 worth anything.



Alternative to CRISP-DM Process Model

Process Model	Who / When	Steps	Best For	Where You See It
CRISP-DM	SPSS and Daimler, 2000	6: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, Deployment	Business projects with a stakeholder and a deadline	The industry default. Most surveyed practitioners still use it
KDD Process	Fayyad, Piatetsky-Shapiro, Smyth, 1996	5: Selection, Preprocessing, Transformation, Data Mining, Interpretation	Research where the goal is knowledge, not a product	Both our textbooks. Academic papers
SEMMA	SAS Institute, 1998	5: Sample, Explore, Modify, Model, Assess	Modeling work inside SAS Enterprise Miner	SAS shops. Banking and insurance
TDSP	Microsoft, 2016	4 looping: Business Understanding, Data Acquisition, Modeling, Deployment	Teams needing roles, git, and shared code structure	Azure projects. Corporate data science teams
OSEMN	Hilary Mason and Chris Wiggins, 2010	5: Obtain, Scrub, Explore, Model, iNterpret	Fast solo analysis. A mental checklist, not a standard	Blogs, bootcamps, informal work
MLOps lifecycle	Emerged around 2018, no single author	Continuous: Build, Deploy, Monitor, Retrain	Models that must keep running after launch	Production ML teams. Extends past CRISP-DM's step 6

KDD: Knowledge Discovery in Databases

Fayyad, Piatetsky-Shapiro and Smyth, 1996

The academic version. The goal is knowledge, not a shipped product.

1.Selection
Choose which data and which variables go into the study.

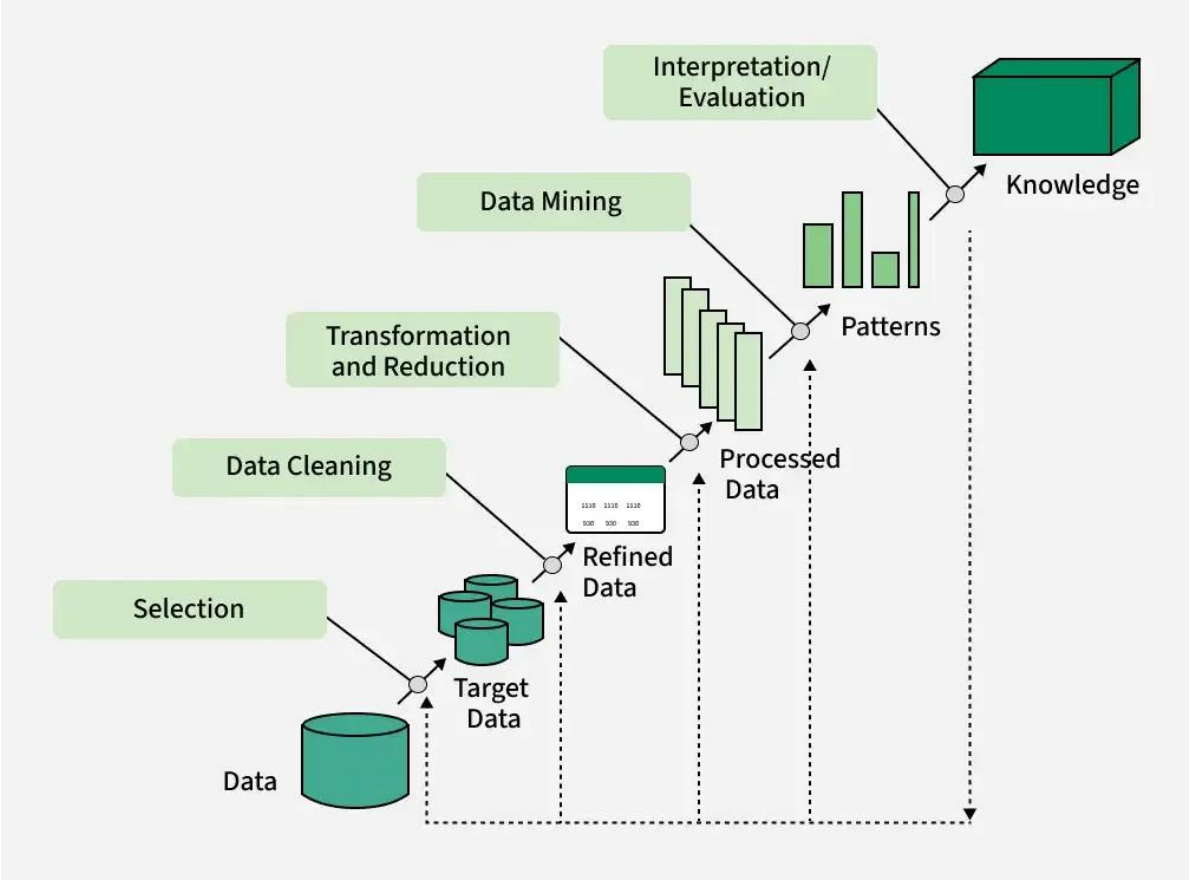
2. Preprocessing
Clean the data. Handle missing values, remove noise, fix errors.

3. Transformation
Reshape the data into the form the algorithm needs. Encode, scale, reduce.

4. Data Mining
Run the algorithm and produce patterns.

5. Interpretation and Evaluation
Judge which patterns are real and useful, then present them as knowledge.

Data mining is step 4 only.
The other four decide whether step 4 was worth anything.



Source: https://www.geeksforgeeks.org/dbms/kdd-process-in-data-mining/

SEMMA Process Model

Sample, Explore, Modify, Model, Assess | SAS Institute, 1998

The academic version. The goal is knowledge, not a shipped product.

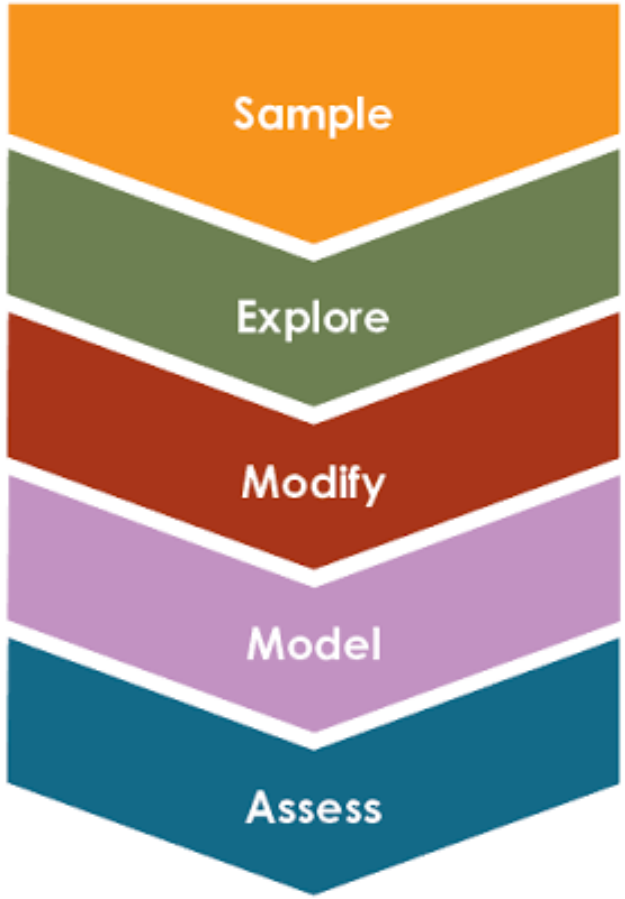
1. Sample
Take a subset large enough to be representative and small enough to work with quickly.

2. Explore
Search the data for patterns, trends, and anomalies to guide what comes next.

3. Modify
Create, transform, and select variables to prepare the data for modeling.

4. Model
Fit algorithms to the prepared data to produce the outcome you want.

5. Assess
Evaluate how well the model performs and how reliable its results are.



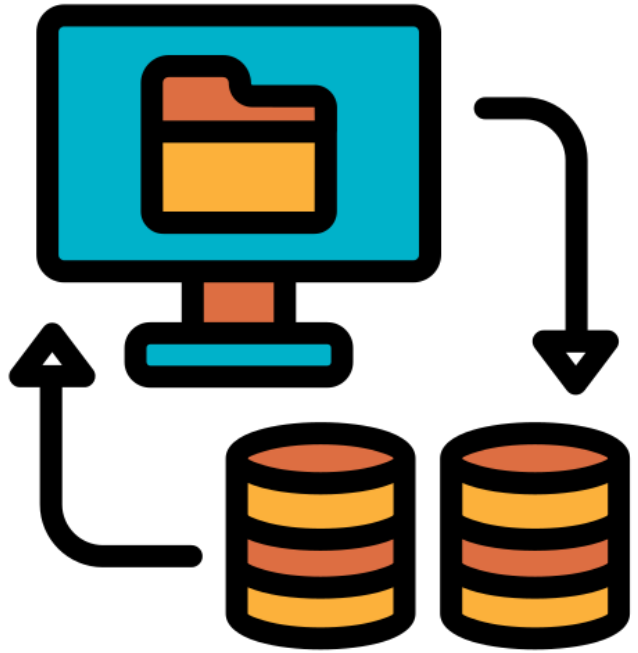
Source: https://www.datascience-pm.com/semma/

Notice what is missing. There is no business understanding step. SEMMA assumes somebody already asked the question.

Our Working Environment

Where to get data?

- UC Irvine Machine Learning Repository
- Kaggle
- data.gov
- Google Dataset Search
- **Your research domain data**



Our Working Environment

Ames Housing Dataset

About Dataset

The Ames Housing Dataset is a well-known dataset in the field of machine learning and data analysis. It contains various features and attributes of residential homes in Ames, Iowa, USA. The dataset is often used for regression tasks, particularly for predicting housing prices.

Here are some key details about the Ames Housing Dataset:

- Number of Instances: The dataset consists of 2,930 instances or observations.
- Number of Features: There are 79 different features or variables that describe various aspects of the residential properties.
- Target Variable: The target variable in the dataset is the "SalePrice," representing the sale price of the houses.
- Data Types: The features include both numerical and categorical variables, covering a wide range of aspects such as lot size, number of rooms, location, construction, and more.

The Ames Housing Dataset is widely used in the machine learning community for tasks such as regression modeling, feature engineering, and predictive analytics related to housing prices. It serves as a valuable resource for developing and testing machine learning algorithms and techniques in the real estate domain.

Our Working Environment

Where to write code?

- **Google Colab**
- Jupyter Notebook
- VS Code
- Kaggle Notebooks

Why we use Google Colab?

- Runs in the browser, nothing to install
- Free GPU when we need it later
- Saves to your Google Drive automatically



Our Working Environment

The Libraries We Use

- **Pandas:** tables and data frames
- **NumPy:** numbers and arrays
- **Matplotlib:** plots
- **Seaborn:** nicer plots, less code
- **scikit-learn:** the algorithms











Run Examples...

Understanding Data

MS SubClass identifies the type of dwelling

20:	1-story, 1946 and newer, all styles
60:	2-story, 1946 and newer
70:	2-story, 1945 and older

All Stored the Exact Same Way (Ames Housing Dataset)

	MS SubClass	Year Built	Overall Qual	SalePrice
	20	2003	7	208500
	60	1976	6	181500
	20	2001	7	223500
	70	1915	7	140000
	60	2000	8	250000
Avg:	**57.4**	**1971.4**	**6.09**	**180,796**

Same data type. Same storage. Same everything.

Run Examples...

Understanding Data

Data Attributes:

A property or characteristic of an object that may vary, either from one object to another or from one time to another.

Introduction to Data Mining, 2nd Edition

Their four types:

1. Nominal

- The values are labels. Nothing more.
- The only fair question: are these two the same, or different?

- **From Ames Dataset:**
  MS SubClass : 60, 70, 80 …
  Roof Style: Gable, Hip, Flat, Mansard
  Central Air: Y or N. Binary nominal

Understanding Data

Data Attributes:

A property or characteristic of an object that may vary, either from one object to another or from one time to another.

Introduction to Data Mining, 2nd Edition

Their four types:

2. Ordinal

- You can rank the values. You cannot subtract them.
- The distance between one rank and the next is unknown.

- **Examples:**

  Kitchen Qual:
  Fireplace Qual:
  Scale: Ex (excellent), Gd (good), TA (typical/average), Fa (fair), Po (poor)

  Overall Qual: 1 to 10
  Overall Cond: 1 to 10

Understanding Data

Data Attributes:

A property or characteristic of an object that may vary, either from one object to another or from one time to another.

Introduction to Data Mining, 2nd Edition

Their four types:

3. Interval

- Differences are meaningful. Ratios are not.
- Zero is just a convention somebody picked.

- **Temperature in Celsius:**
  30C minus 20C is the same gap as 20C minus 10C. Fine.
  30C is NOT twice as hot as 15C. Zero Celsius is arbitrary.

- **From Ames Dataset:**
  Year Built :original construction year

Understanding Data

Data Attributes:

A property or characteristic of an object that may vary, either from one object to another or from one time to another.

Introduction to Data Mining, 2nd Edition

Their four types:

4. Ratio

- Zero means none of the thing. It is not a convention.
- Multiplication and division both make sense.

- **From Ames Dataset:**
  Lot Area: lot size in square feet
  Gr Liv Area: above-grade living area
  SalePrice: the target

- Most measurements you meet are ratio. Height, weight, count, money.

Understanding Data

Data Attributes:

A property or characteristic of an object that may vary, either from one object to another or from one time to another.

Introduction to Data Mining, 2nd Edition

Column	Type	What it means	Legal average?
MS SubClass	Nominal	Labels only. No order at all	No
Overall Qual	Ordinal	Ranked, but the gaps are unknown	No
Year Built	Interval	Gaps are real, zero is arbitrary	Yes
SalePrice	Ratio	True zero, so ratios work	Yes

Understanding Data

What You Are Allowed to Compute

Operation	Nominal	Ordinal	Interval	Ratio
Equality (= , !=)	Yes	Yes	Yes	Yes
Order (< , >)	No	Yes	Yes	Yes
Difference (-)	No	No	Yes	Yes
Ratio (/)	No	No	No	Yes
Mode	Yes	Yes	Yes	Yes
Median	No	Yes	Yes	Yes
Mean	No	No	Yes	Yes

Understanding Data

Discrete or Continuous?

- Discrete: countable values, with gaps between them.
- Continuous: any value in a range, limited only by your instrument.

- **From Ames Dataset :**

  Discrete (counted)
  Bedroom AbvGr: bedrooms above grade
  Full Bath, Half Bath: full and half bathrooms
  Bsmt Full Bath, Bsmt Half Bath
  Kitchen AbvGr: kitchens

  Continuous (measured)
  Lot Area: lot size in square feet
  Lot Frontage: feet of street frontage
  Gr Liv Area: above-grade living area

Understanding Data

Which Type Points to Which Task

Your target column	Type	The task
Central Air (Y/N)	Nominal, binary	Classification
Kitchen Qual (Ex to Po)	Ordinal	Classification, ordered
SalePrice ($)	Ratio	Regression
No target at all	—	Clustering
Rows that do not fit	—	Anomaly detection
Features that appear together	—	Association

Understanding Data

Five Things Wrong With Real Data

Missing Values

- Missing at random: a sensor blipped. No pattern to it.
- Missing for a visible reason: guest checkouts have no Customer ID.
- Missing for a hidden reason: high earners skip the income question.

- **Only the first one is safe to simply fill in.**

Understanding Data

Five Things Wrong With Real Data

Duplicates, Outliers and Noise

- Duplicates: the same record entered twice. Inflates every count.
- Outliers: a real value far from the rest. Might be your best row.
- Noise: a wrong value. Measurement error, or somebody's typo.

- **Outliers and noise look identical in a plot.**
- Only domain knowledge tells them apart.

Understanding Data

Five Things Wrong With Real Data

Inconsistency

- Same real thing, written several ways. Your code sees several things.
- **Country column, all one country:**
  United Kingdom / UK / U.K. / uk / united kingdom
  Five separate groups in your groupby. One actual country.
- Units too. Price in pounds or pence? Weight in kg or lb?
- **Nothing errors out. Your numbers are just wrong.**

Run Examples...

What EDA is Actually For

Definition:

- Exploratory Data Analysis (EDA) is the preliminary investigation of data to understand its characteristics, find its problems, and decide what to do next

- **Three questions you are answering:**
  - What shape is each column?
  - Where are the problems?
  - What relates to what?

EDA is how you find out your data is broken.
Before your model finds out and lies to you instead.

What EDA is Actually For

Match the Plot to the Attribute Type

What you want to see	Plot	Works on
One column, its distribution	Histogram	Interval, Ratio
One column, counts per category	Bar chart	Nominal, Ordinal
One column, spread and outliers	Boxplot	Interval, Ratio
Two columns, a relationship	Scatter plot	Interval, Ratio
A category against a number	Grouped boxplot	Nominal plus Ratio
Where the blanks are	Heatmap of nulls	Anything

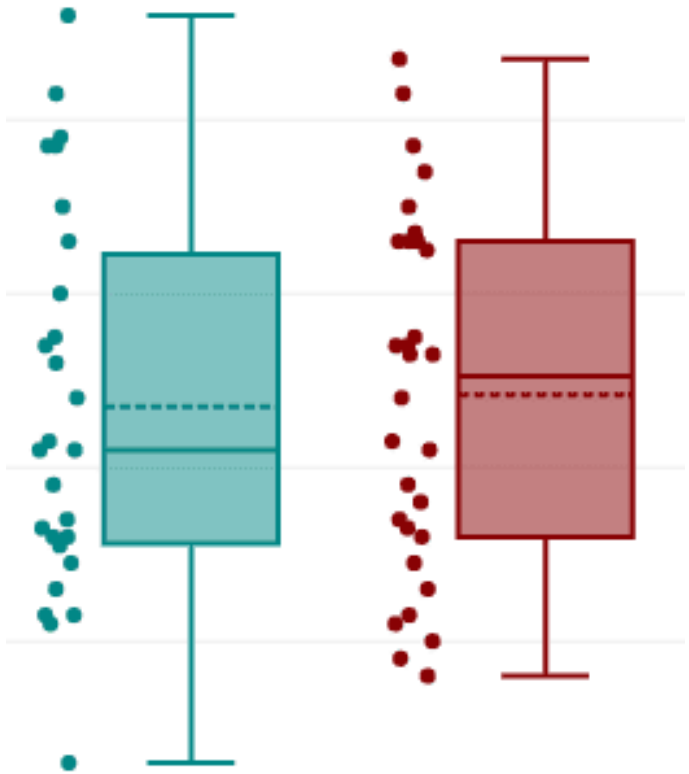
Pick the plot from the column type. Not from what looks nice.

What EDA is Actually For

Boxplot: The One Plot You Must Know

- The box holds the middle half of your data.
- The line inside the box is the median, not the mean.
- The whiskers reach out 1.5 times the width of the box.
- Dots past the whiskers are flagged as possible outliers.

- **Flagged is not the same as wrong.**
- A boxplot makes a suggestion. You make the decision.



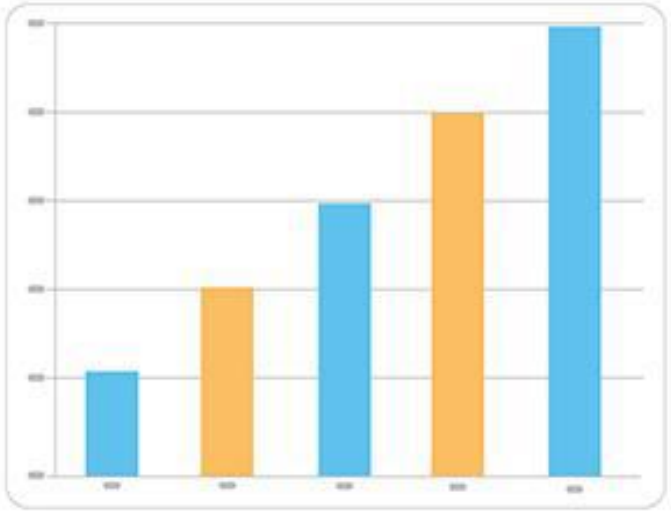
What EDA is Actually For

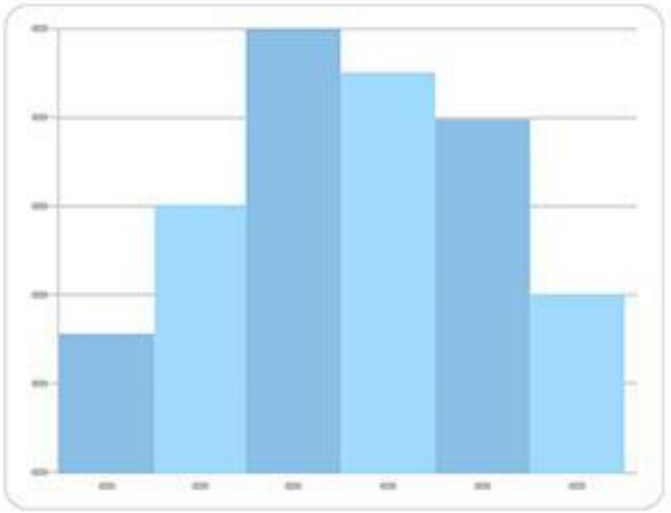
Histogram vs Bar Chart: Not the Same Plot

- Histogram: one numeric column, split into bins.
- Bar chart: one categorical column, one bar per category.
- Histogram bars touch. Bar chart bars have gaps.
- The x-axis of a histogram has an order. A bar chart usually does not.

- **Use a histogram on Interval or Ratio. Use a bar chart on Nominal or Ordinal.**

- Histogram shows you shape. Bar chart shows you counts.



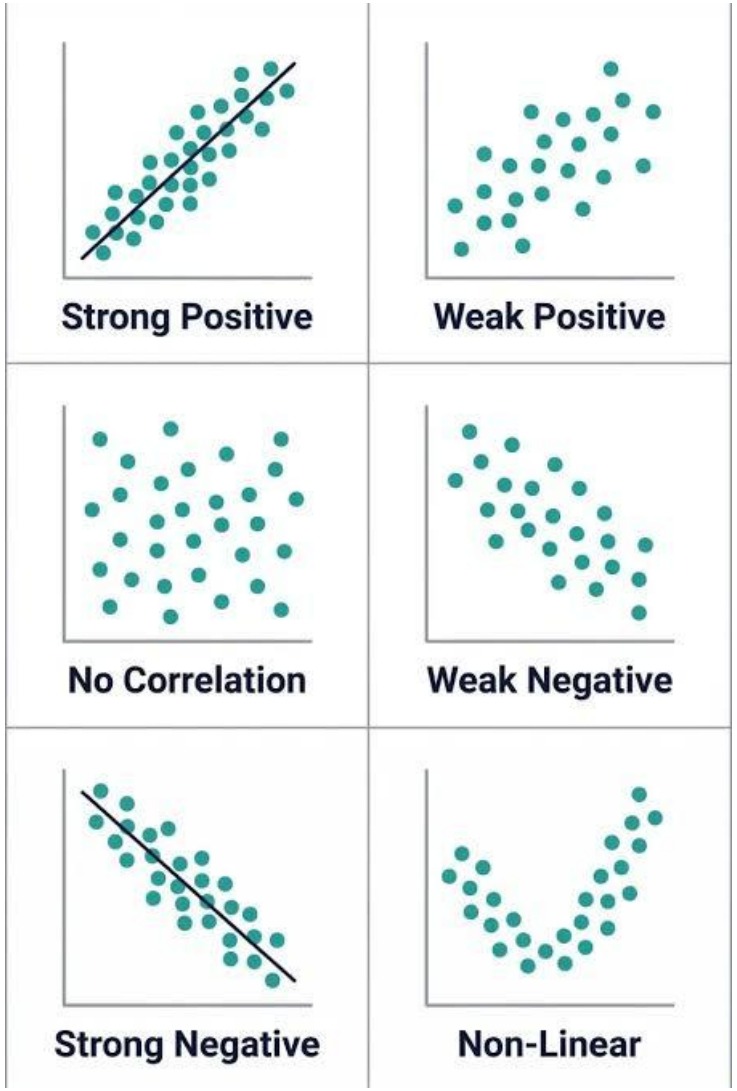


What EDA is Actually For

Scatter Plot: Two Columns at Once

- One dot per row. Two numeric columns, one on each axis.
- You are looking for a shape, not individual points.
- Upward cloud means they rise together. Flat cloud means no relation.
- The points far from the cloud are your outliers.

- A pattern here is a hypothesis, not a finding.



What EDA is Actually For

Correlation Heatmap: What Moves With What

- Every numeric column against every other numeric column.
- Value runs from -1 to +1. Zero means no linear relationship.
- Red means they rise together. Blue means one rises as the other falls.
- The diagonal is always 1. Every column matches itself.

- Correlation is not causation. e.g., Ice cream and drowning.

